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A global cyber-threat intelligence system with artificial intelligence and convolutional neural network

Fahim Sufi

Monash University, Melbourne, VIC 3004, Australia

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ABSTRACT

Global cyber-attacks significantly impact the economy, society, organizations, and individuals. Existing research on cyber-attacks lacks in demonstrating Artificial Intelligence (AI) based analytical solutions for providing country-wide cyber threat intelligence. Cyber strategists at a national level require AI-based decision support systems for deciding a country's cyber posture or preparedness. This paper proposes an AI-based solution that autonomously collects multidimensional cyber-attack data on social media posts on cyber-related outcry. The proposed system provides critical analytical capability in the cyber-threat spectrum and uses sophisticated AI-based algorithms for anomaly detection, prediction, sentiment analysis, location detection, translation, etc. The proposed system was deployed from 11 October 2022 to 31 October 2022. During these 21 days, the system autonomously collected 30,203 records on cyber threats covering multiple dimensions of cyber-threat. These dimensions included country-wide daily cyber-attack records by ransomware, exploits, web threats, spam, malicious mail, network attacks, local infections, and on-demand scan. Moreover, the system performed AI-based acquisition and analysis of 3789 cyber-related tweets from 3402 tweet users in 37 different languages. The system also autonomously translated 893 non-English tweets. The proposed system is the first solution that uses Convolutional Neural Network (CNN) based anomaly detection to detect abnormalities in cyber-threat spectrum worldwide and predict cyber-attacks automatically. The proposed system was demonstrated to provide evidence-based decisions on global cyber threats in multiple platforms, including iOS, Android, and Windows.

1. Introduction

Worldwide Cyber-attacks had costed almost 1 trillion USD to global economy in 2020 [1]. Cybercrimes are projected to cost as much as 10.5 trillion USD annually by 2025 [2]. Larger economies like China, Brazil, United States, India, Mexico, France, Australia and even United Arab Emirates suffer from billions of dollars in consumer loss through cybercrimes [3]. For example, China's consumer loss through cyber-crime was 66.3 billion USD in 2017 [3]. Other than creating economic losses, Cyber-attacks has detrimental effect at social and psychological level touching the lives of human being [4]. For example, very recently one of Australia's telecom giant, Optus have been exposed to data breach as a result of cyber-attack, which has created massive stress and anger on consumers [5,6]. Because of this cyber-attack, information of Optus customers (e.g., name, date of birth, email addresses, driver's license, Medicare card, passport numbers etc.) have been exposed causing distress among 2.1 million consumers [6,7]. Most recently, sensitive patient information that include medical diagnosis & procedure were stolen by cyber criminals from Australia's number 1 health insurer [8]. Hence, cyber-crime is one of the most critical concerns of modern nations, states, organizations, and individuals.

To reduce the impact of cyber-crimes two critical mandates are aggregation of cyber related data and sophisticated analytical algorithms to identify and deter threats. The requirements of cyber data have been portrayed in recent research works [1,9]. On the other hand, the requirement of Artificial Intelligence (AI) based sophisticated analytical algorithms have been demonstrated in [10–19]. While AI can be used to conduct sophisticated cyber-attacks [10,11], AI can also be used in detecting cyber-attack as demonstrated in [11–19]. However, none of these existing literatures provide country-level historical cyber statistics, country level cyber prediction, and anomaly detections with AI. Therefore, the proposed cyber intelligence dashboard contributes towards the following Critical Requirements (CR) to cyber strategists for making evidence-based decisions on national cyber posture:

CR 1: Analyze threat spectrum against different dimensions of cyber-attacks for a particular country and compare with other countries in the globe

CR 2: Analyze changes of threat spectrum by time for any countries (e.g., daily, weekly, monthly change)

CR 3: Predict cyber-attacks for any countries in the globe by analyzing past cyber-attack data

E-mail address: fahim.sufi@monash.edu.

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Fig. 1. Aggregation of multidimensional cyber-attack data and cyber related social media data.

CR 4: Detect Anomalies in cyber-threat spectrums for any countries in the globe by analyzing past cyber-attack data

CR 5: Analyze the view of social media users on various cyber issues over time

CR 6: Obtain the above analytical requirements (i.e., CR 1 to CR 5) in any platforms (i.e., Windows, Android, iOS) on any devices (e.g., mobile, tablet, desktop)

In this paper, we propose an innovative solution that successfully meets the analytical requirements highlighted within CR 1 to CR 6, by aggregating cyber related data from multiple sources and using AI-based services and algorithms. As seen from Fig. 1, multidimensional Cyber-attack data were obtained from leading Anti-virus software vendor (e.g., Kaspersky [20]) and Cyber related social media data was obtained from Twitter. As seen from Fig. 1, the multidimensional cyber-attack statistics included Ransomware [21], Exploits [22], Web threats [23], Spam [24], Malicious mail [25], Network attacks [26], Local Infections [27], and On-demand scan [28] for all the countries in the globe.

2. Background literature

Existing literatures in Cyber-Threat intelligence mainly focused on an organizational perspective, where cyber-attack impacted an individual organization for financial or political benefit by the perpetrator [12–19]. While these perspective serves the concerns of an individual organizational entity, this does not cater the need of political heads of states having a wholistic view on their nation. National level concern on cyber-attack requires country-wide cyber threat intelligence that includes attack spectrum covering an individual country along with other countries in the globe for benchmarking purpose. Using this wholistic view of all the cyber-attack statistics for a country in comparison with other countries allows the political decision maker to determine their posture on cyber defense. For example, if Australia is facing unusually higher number of cyber-attacks compared to other nations like New Zealand, United Arab Emirates, Malaysia, then Australia should place their Cyber defense posture at a higher level. Country-wide cyber threat intelligence is critical for a nation's Cyber Preparedness.

Existing research also suffers from not utilizing AI-based fully automated data acquisition techniques like Web Scraping, Entity Detection for Location Intelligence, Sentiment Analysis as their data requirements were primarily focused on intrusion detections within the network. As a result, most of the existing studies as demonstrated in [12,13,16,17,29], were only obtaining Network Traffic Data. Study in [18] used simulated sensor data on cyber physical system. On the other hand, researcher in [30] used survey data collected from 294 participants. None of these existing research works, obtained cyber threat data from multiple sources like Cyber-attack statistics for all the countries from Anti-Virus Software vendors or Cyber related outcry from live social media posts. Social media data on geopolitical relevance requires AI-based data acquisition and pre-processing as demonstrated in our recent papers [31–38]. As social media contents were fetched, our novel methodology demonstrated in [31–38], applied AI-based Named Entity Recognition (NER) for extracting geospatial intelligence, sentiment analysis for extracting subjective relevance of the content, dynamic translation to understand social media contents in 110 different languages.

Existing literature demonstrates that there have been a growing interest using AI-based Anomaly detection and other deep learning methodologies for detecting cyber-attack [11–19]. However, within these studies Anomaly detection has been used on Network Traffic data for detecting intrusions. Anomaly detection algorithms have never been reported to be used on country-wide cyber-attack statistics.

Finally, almost none of the existing literature on cyber-attack provided interactive dashboards that updates itself automatically for providing evidence-based decisions on cyber posture for strategic leaders of a nation. Since a strategic decision maker needs to make instant decisions, the dashboards are required to be available in a wide range of platforms that includes mobile, tablet or traditional desktops. Existing research work on Cyber-attack depicted in [11–19] were not available a wide range of platforms like Windows, iOS, or Android. It should be mentioned that the requirement of strategic decision makers to make instant decision on a wide range of platforms like mobile phone running iOS or Android have been demonstrated in [31–37,39–43]. Table 1, summarizes the bottleneck of existing studies and clearly demonstrates how the solution presented in this paper aims to address these shortcomings.

Table 1
Literature review of existing studies in Cyber threat analysis with supported features.

Reference	1. Country-Wide cyber threat intelligence	2. Automated AI-Based data acquisition	3. Use of multi-source data	4. AI-based deep learning/ anomaly detection/ prediction	5. Self-updating dashboard	6. Interactive dashboard for decision making	7. Dashboard availability in multiple platforms
[12]	No	No	No. Network Traffic Data	Yes	No	No	No
[13]	No	No	No. Network Traffic Data	Yes	No	No	No
[29]	No	No	No. Network Traffic Data	Yes	No	No	No
[16]	No	No	No	Yes	Yes	No	No
[17]	No	No	No	Yes	No	No	No
[18]	No	No	No. Simulated Sensor Data	Yes	No	No	No
[19]	No	No	No. Network Traffic Data	No	Yes	Yes	No
[30]	No	No	No. Survey Data	No. Only Statistical Methods	No	No	No
Proposed	Yes	Yes	Yes	Yes	Yes	Yes	Yes

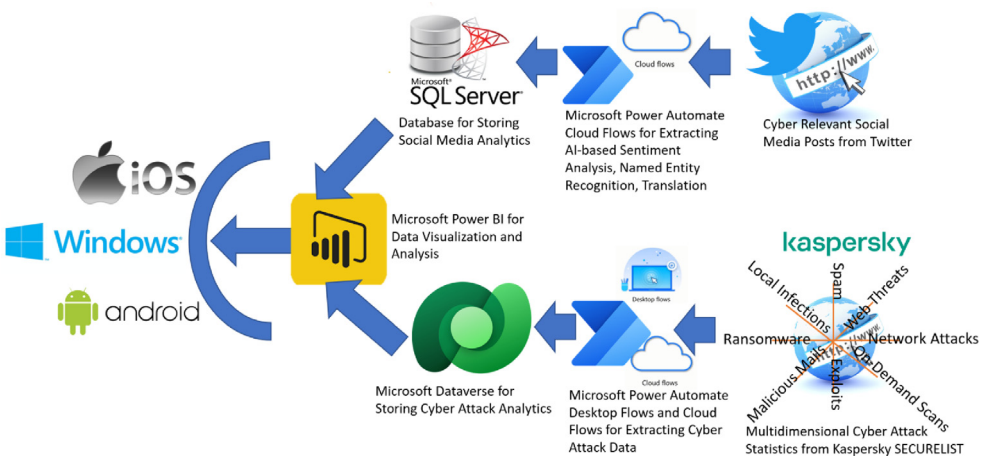


Fig. 2. Architecture of acquiring Multidimensional Cyber Attack data and Cyber relevant social media data for rich data visualization and analytics.

As seen from Table 1, the proposed approach presented in this paper automatically acquires multidimensional data on cyber-attack covering dimensions such as Spam, Ransomware, Web Threats, Network Attacks, Malicious Mails, Local Infections, Exploits, and On Demand Scans. Moreover, the presented approach obtains cyber related posts from live social media from Twitter for having a comprehensive understanding of cyber threats that social media users are talking about. Furthermore, this paper applies AI-based deep learning techniques (i.e., Anomaly Detection) on global Country-Wide attack data to automatically identify abnormally high or low level of attacks on a nation. Finally, the presented solution is deployed into multiple platforms like desktop, tablet, and mobile in a wide range of environments like iOS, Android, Windows. Hence, strategic decision-makers can make instant evidence-based decisions right from their device of choice. The fully interactive dashboard updates (i.e., refreshes itself) providing the most up-to-date information supporting data driven decision support required by today's strategic leaders [44].

3. Material & methods

Multidimensional Global Cyber Attack data is acquired using a combination of Desktop Flow and Cloud Flow in Power Automate [45]. The Desktop flow makes use of unattended Power Automate Desktop flow for automating a series of tasks. This automation opens up browser, goes to 8 different links [21–28] downloads 8 different attack statistics files, and saves these files (as Microsoft Excel .xlsx) into designated One Drive for business folder. With every new file creation, a trigger initiates a separate Microsoft Power Automate cloud flow that processes these attack statistics and stores them in Microsoft Dataverse [46]. Thus, by using Desktop flow and Cloud Flow, the presented system automatically and anonymously updates itself on a daily schedule with daily cyber statistics provided by [21–28]. It should be noted that the Kaspersky's sites in [21–28] does not provide access to historical data. Every day, the daily cyber statistics are replaced with updated statistics. Hence, the presented system builds historical cyber statistics on a daily basis (i.e., not in real-time). However, the presented system would support real-time updates (using the same technology stack of Microsoft Power Automate), when the cyber-attack statistics are provided in real-time (by the data sources).

On the other hand, social media posts from Twitter are extracted using Microsoft Power Automate flow. During this extractions AI-based Sentiment Analysis, NER, Translation is performed as depicted in our most recent research [31–38,42]. Finally, these records are stored in Microsoft SQL Server Database.

Microsoft Power BI [47] obtains both these data sources (i.e., multidimensional cyber threat data of Kaspersky and cyber related social media posts from Twitter) to provide data analytics and data visualizations for the strategic decision makers in iOS, Windows, and Android platforms. Fig. 2 depicts this overall process. Furthermore, Table 2 provide details into various technology components used within the proposed system along with justifications. Table 2 refers to individual features that was provided within Table 1 and maps those features to technology components.

As it appears from Fig. 2 and Table 1, AI-based anomaly detection is the key technology empowering the decision analytics. The anomaly detection adds additional feature to line charts by automatically detecting anomalies within time-series data. It also provides Natural Language Processing (NLP) based [48] explanations for the anomalies facilitating root-cause analysis. In our most recent study, we have used AI-based anomaly detection for identifying abnormal cases of

Table 2
Technology components used for automation.

Technology component	Purpose	Supported feature of Table 1
Microsoft Power Automate	• Capturing Tweets related to Cyber • AI-based Translation • AI-based Sentiment Analysis • AI-based Named Entity Recognition	• Feature 1 (Country-Wide) • Feature 2 (data acquisition) • Feature 3 (multisource data)
Microsoft Desktop Automate	• Unattended Robotic Process Automation for acquiring Cyber-attack data from 8 different sources and saving them as .xlsx files within Microsoft OneDrive	• Feature 1 (Country-Wide) • Feature 2 (data acquisition) • Feature 3 (multisource data)
Microsoft Power BI	• AI-Based Anomaly Detection • Exponential Smoothing Prediction • Dashboard for Windows • iOS App • Android App	• Feature 4 (Use of AI/CNN) • Feature 5 (Dashboard) • Feature 6 (Dashboard) • Feature 7 (Multi-Platform)
Microsoft Dataverse	• Storage of Daily Cyber Attack Statistics	• Feature 1 (Country-Wide) • Feature 2 (data acquisition)
Microsoft SQL Server	• Storage & Management of Tweets	• Feature 1 (Country-Wide) • Feature 2 (data acquisition)

landslides along with the root causes [39,40], abnormalities of disaster events from social media posts [36,49], anomalies on global events by monitoring 2397 global news sources [31,32] and even on COVID-19 situational awareness [43]. Before delving into the details of anomaly detection, we present the problem definition.

Problem 1. Given a sequence of real values, that is, $x = x_1, x_2, x_3, \ldots, x_n$, the task of time-series anomaly detection is to produce an output sequence $y = y_1, y_2, y, \ldots, y_n$, where $y_i \in \{0, 1\}$ denotes whether x_i is an anomaly point.

The implemented solution borrowed the Spectral Residual (SR) from the visual saliency detection domain and then applied a Convolutional Neural Network (CNN) to the results produced by the SR model [50]. The SR algorithm consists of three major steps:

1. Perform Fourier transform to obtain the log amplitude spectrum.
2. Calculate the SR.
3. Perform inverse Fourier transform, which transforms the sequence back to the spatial domain.

$$A(f) = Amplitude(f(x)) \tag{1}$$

$$P(f) = Phrase(f(x)) \tag{2}$$

$$L(f) = log(A(f)) \tag{3}$$

$$AL(f) = h_q(f).L(f) \tag{4}$$

$$R(f) = L(f) - AL(f) \tag{5}$$

$$S(x) = \left\| f^{-1}(\exp(R(f) + iP(f))) \right\| \tag{6}$$

where f and f^1 denote the Fourier transform and inverse Fourier transform, respectively; x is the input sequence with shape nX1; $A(f)$ is the amplitude spectrum of sequence **x**; $P(f)$ is the corresponding phase spectrum of sequence x; $L(f)$ is the log representation of $A(f)$; and $AL(f)$ is the average spectrum of $L(f)$, which can be approximated by convoluting the input sequence by $h_q(f)$, where $h_q(f)$ is a $q \times q$ matrix defined as:

$$h_q(f) = \frac{1}{q^2}\begin{bmatrix} 1 & 1 & \ldots 1 \\ 1 & 1 & \ldots 1 \\ \ldots & \vdots & \ddots 1 \\ 1 & 1\ldots & 1 \end{bmatrix} \tag{7}$$

$R(f)$ is the SR, that is, the log spectrum $L(f)$ minus the averaged log spectrum $AL(f)$. The SR serves as a compressed representation of the sequence, whereas the innovation part of the original sequence becomes more significant. Last, the sequence was transferred back to the spatial domain using an inverse Fourier transform. The resultant sequence $S(\mathbf{x})$ is referred to as the saliency map [51]. The values of the anomaly points are calculated as follows:

$$x = \left(\overline{x} + mean\right)(1 + var).r + x \tag{8}$$

where $\overline{x}$ is the local average of the preceding points, *mean* and *var* are the mean and variance of all points within the current sliding window, and $r \sim N\ (0,\ 1)$ is randomly sampled. In this process, CNN is applied to the saliency map instead of to the raw input, thus increasing the efficiency of the overall process of anomaly detection [50,51]. In fact, we implement anomaly detection in three steps using Microsoft Power BI's Line Chart visualization [31,32,36,39,40,43,49]:

- Detect all the anomalies within the time series (i.e., any values that lie outside the threshold range). For the anomaly detection process, all the country specific attack-statistics (i.e., time-series data) was used. Microsoft Power BI's line chart visualization uses CNN (including Fourier transform, SR, and inverse Fourier Transform) to perform anomaly detection on the country specific attack statistics.
- Identify the main drivers of these anomalies. To obtain the main drivers both social-media data and cyber-attack statistics from Kaspersky (i.e., [20]) were utilized.
- Explain the results in a natural language (explanation of the root cause) using NLP [48].

It should be noted that the above process of CNN-based anomaly detection is almost instantaneous with Microsoft BI's implementation [52].

As seen from Table 2, the presented system uses an exponential smoothing algorithm for predicting country-level cyber-attacks. Exponential smoothing is a prominent and adept time series forecasting technique, adroitly adapted to the realm of cyber-attack prediction [53,54]. Through the artful interplay of weighted averages, this algorithm discerningly identifies trends and latent patterns embedded within historical cyber-attack data. By iteratively recalculating forecasts and adeptly modulating the smoothing parameter, the algorithm gracefully adjusts its predictive capacity to accommodate temporal precedence while attenuating the influence of historical data [55]. With due cognizance of data quality and quantity, and the steadfastness of underlying cyber-attack patterns, the efficacy of this method (i.e., exponential smoothing) aspires to illuminate the future landscape of cyber threats with precision and erudition.

4. Results

The proposed methodology was deployed and implemented using different components of Microsoft Power Platform [45–47] as described

Table 3
Number of daily multidimensional cyber- threat statistics downloaded from 11 October 2022 till 30 October 2022 using the proposed system.

	Exploit	Local infection	Malicious mail	Network attack	On-Demand scan	Ransomware	Spam	Web threat	Count of attack statistics
Tuesday, 11 October 2022	185	194	185	186	194	142	161	194	**1441**
Wednesday, 12 October 2022	185	194	185	186	194	142	161	194	**1441**
Thursday, 13 October 2022	183	194	188	184	194	147	155	194	**1439**
Friday, 14 October 2022	178	194	191	185	194	151	159	194	**1446**
Saturday, 15 October 2022	185	194	189	185	194	149	162	194	**1452**
Sunday, 16 October 2022	181	194	186	184	194	139	162	194	**1434**
Monday, 17 October 2022	181	194	174	181	194	130	177	194	**1425**
Tuesday, 18 October 2022	180	194	173	173	193	128	176	194	**1411**
Wednesday, 19 October 2022	184	194	191	187	194	146	176	194	**1466**
Thursday, 20 October 2022	184	194	191	187	194	146	176	194	**1466**
Friday, 21 October 2022	179	194	190	185	194	150	179	194	**1465**
Saturday, 22 October 2022	181	194	188	184	193	143	169	194	**1446**
Sunday, 23 October 2022	177	194	182	183	194	134	163	194	**1421**
Monday, 24 October 2022	176	194	170	178	194	138	163	193	**1406**
Tuesday, 25 October 2022	178	194	166	175	193	124	167	194	**1391**
Wednesday, 26 October 2022	183	194	188	187	194	144	175	194	**1459**
Thursday, 27 October 2022	181	194	193	184	194	142	174	194	**1456**
Friday, 28 October 2022	182	194	186	188	194	146	160	194	**1444**
Saturday, 29 October 2022	184	194	187	186	194	144	156	194	**1439**
Sunday, 30 October 2022	185	194	186	186	194	143	168	194	**1450**
Monday, 31 October 2022	183	193	175	179	194	129	158	194	**1405**
Total	**3815**	**4073**	**3864**	**3853**	**4071**	**2957**	**3497**	**4073**	**30 203**

Table 4
Number of daily cyber related tweets captured and processed with AI services from 13 October 2022 till 31 October 2022 using the proposed system.

Date	No. of twitter ID	No. of user ID	No. of locations	No. of tweet languages	Retweets	Average negative sentiments	Average neutral sentiments	Average of positive sentiments	No. of translations
Thursday, 13 October 2022	52	51	29	7	80 707	0.398	0.422	0.180	12
Friday, 14 October 2022	211	189	122	15	77 089	0.399	0.402	0.199	67
Saturday, 15 October 2022	219	208	116	18	408 635	0.292	0.461	0.247	74
Sunday, 16 October 2022	208	205	111	18	428 407	0.312	0.442	0.246	67
Monday, 17 October 2022	221	208	122	14	188 791	0.295	0.464	0.240	60
Tuesday, 18 October 2022	186	180	101	18	49 255	0.315	0.492	0.193	56
Wednesday, 19 October 2022	226	219	133	18	132 222	0.354	0.421	0.225	55
Thursday, 20 October 2022	216	215	123	17	231 915	0.321	0.469	0.210	51
Friday, 21 October 2022	206	204	129	17	533 082	0.434	0.413	0.152	37
Saturday, 22 October 2022	219	209	118	14	134 067	0.409	0.401	0.190	46
Sunday, 23 October 2022	223	207	116	18	34 249	0.333	0.466	0.200	69
Monday, 24 October 2022	226	218	128	16	88 944	0.440	0.353	0.206	59
Tuesday, 25 October 2022	227	219	118	20	200 700	0.434	0.403	0.164	46
Wednesday, 26 October 2022	219	205	113	13	30 097	0.375	0.413	0.211	48
Thursday, 27 October 2022	222	219	121	14	175 143	0.339	0.423	0.238	47
Friday, 28 October 2022	218	212	124	14	287 112	0.388	0.377	0.235	48
Saturday, 29 October 2022	224	215	126	14	176 450	0.414	0.356	0.230	41
Sunday, 30 October 2022	222	215	114	12	217 949	0.351	0.457	0.208	48
Monday, 31 October 2022	209	205	113	18	252 942	0.322	0.480	0.199	55
Total	**3998**	**3607**	**1643**	**38**	**3 727 756**	**0.363**	**0.426**	**0.210**	**941**

in the previous section from 11 October 2022 till 31 October 2022. During this time, 30,203 records on global cyber-attack statistics covering exploit, local infection, malicious mail, network attack, on-demand scan, ransomware, spam and web threat were automatically captured using Microsoft Power Automate and Microsoft Dataverse. Table 3 provides detailed statistics on these 30K records. Each of the columns Table 3, starting from Exploit to Web Threat daily statistics automatically fetched by the presented system from Kaspersky's threat statistics URL (i.e., https://statistics.securelist.com/) for 8 different types of cyber threats (i.e., Exploit, Local Infection, Malicious Mail, Network Attack, On-Demand Scan, Ransomware, Span, and Web Threat). The Daily aggregation of these 8 types of threats is provided in "Count of Attack Statistics" column of Table 3.

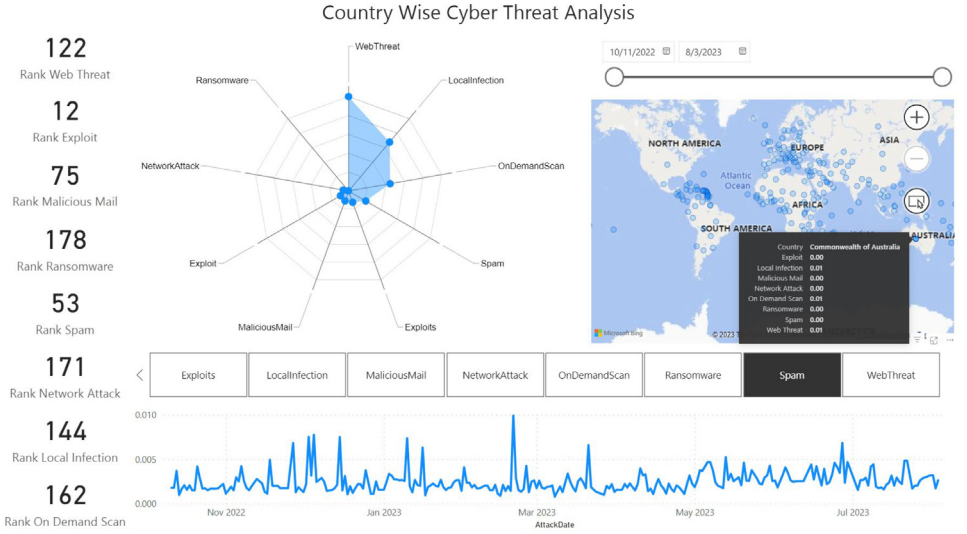
Moreover, our implementation with Microsoft Power Automate and Microsoft SQL Server, automatically collected 3789 Tweets from 3402 distinct Tweet users. As seen from Table 4, these tweets were automatically processed with AI-based algorithms to comprehend Tweets in 37 distinct languages. 893 Non-English Tweets were also translated dynamically. The column "No. of Translations" shows how many non-English Tweets were captured each day with subsequent translation and analysis (with sentiment analysis). This column (i.e., number of translation) signifies the capability and capacity of the presented system in comprehending multi-lingual tweets for a more comprehensive situational awareness. Detailed sentiment analysis revealed that the highest negative sentiment (with an average negative sentiment of 0.44) were noted on 24 October 2022, when many people were upset about recent cyber-attacks on Australia's telecom and health insurance providers.

While Table 3 shows the statistics pertaining to cyber-attack data, Table 4 demonstrates statistics on the analyzed social-media data. The presented system started obtaining social media data from 13 October 2022 as opposed to 11 October 2022. Hence, the start date of Table 3 and Table 3 differs by 2 days. It should be noted that cyber related information (either real attack data or data obtained through social media) changes on a day-to-day basis depending geo-political, social, military, and diplomatic dynamics. These changing dynamics of cyber related data provides completely different AI-driven insights as reported in recent literatures [56–59].

The first requirement as identified in CR 1 is to analyze threat spectrum against different dimensions of cyber-attacks for a particular



(a) Cyber Threat spectrum selected for Bangladesh



(b) Cyber Threat spectrum selected for Australia

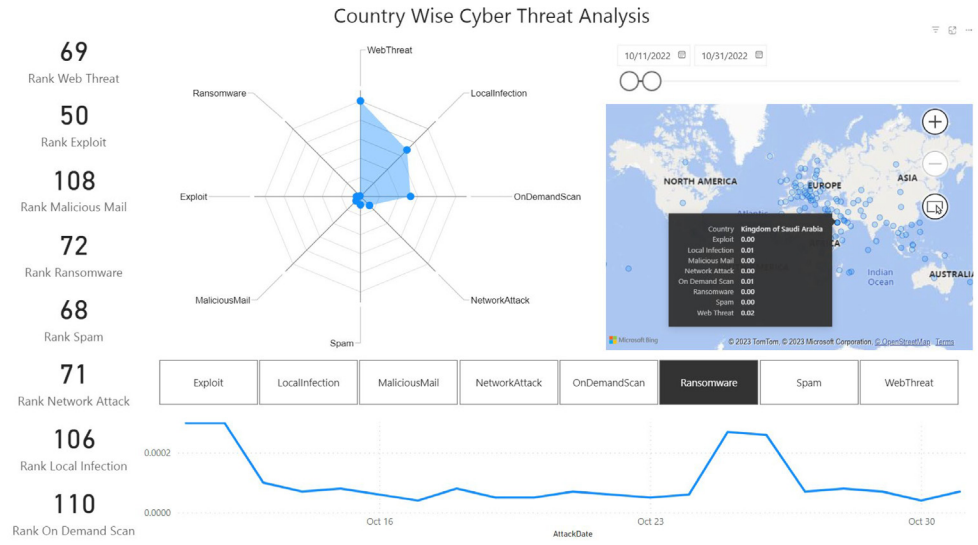
Fig. 3. Analyzing cyber threat spectrum against multiple dimensions of cyber-attacks between Bangladesh and Australia fulfilling CR 1.

country and compare with other countries in the globe. In Fig. 1(a), as Bangladesh is highlighted, the strategic decision maker can easily observe the ranking on various cyber threats (left side). For Bangladesh, during the monitored period, Web threat rank was 25, Exploit rank was 46, Malicious mail rank was 19, Ransomware rank was 8, Spam Rank was 50, Network attack rank was 7, Local Infection rank was 8 and finally, On-Demand scan rank was 4 on a global scale. Also, the radar chart in the center of Fig. 3(a) informs that within Bangladesh percentage of location infection rate is higher followed by On-Demand scan, Web Threat, Network attack and then other dimensions. Fig. 3(b) demonstrates the cyber threat spectrum for Australia, is quite different during that same time period. In Australia (i.e., Fig. 3(b)), the percentage of Web Threat is highest, followed by location infections, On-Demand Scan, Span, Exploit and then others. Fig. 3 provides an example of how a strategic decision-maker can the analyze difference in cyber threat spectrums among a number of countries.

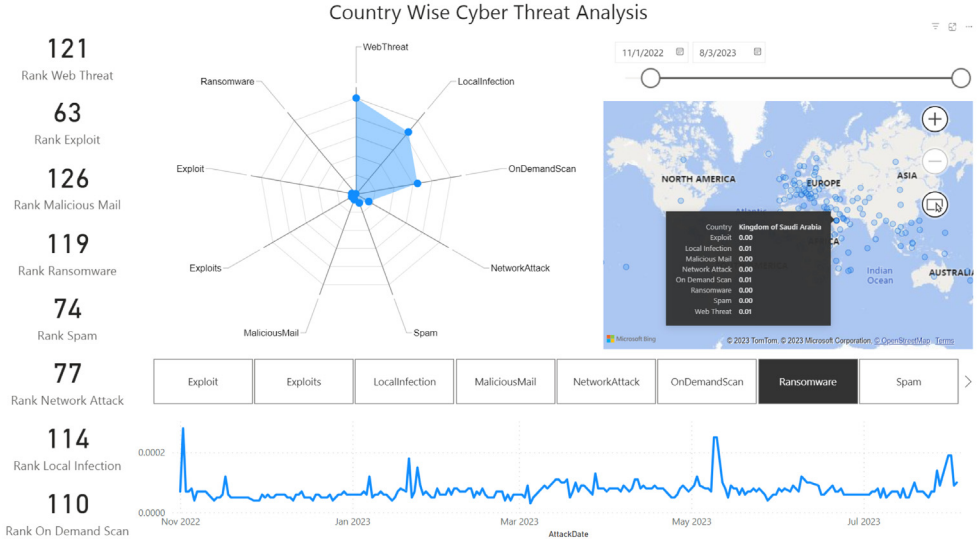
Then, CR 2 mandates analyzing change of threat spectrum by time for any countries. Fig. 4 demonstrates the cyber threat spectrum for Saudi Arabia between two different timeframes. As seen from Fig. 4(a), Saudi Arabia's ranking on web threat, exploit, malicious mail, ransomware, spam, network attack, local infection, and on-demand scan were 69, 50, 108, 72, 68, 71, 106, and 110 respectively from 11 October 2022 till 31 October 2022. However, from 1 November 2022 to 3 August 2023, the global ranking of Saudi Arabia in terms of web threat, exploit, malicious mail, ransomware, spam, network attack, local infection, and on-demand scan were 121, 63, 126, 119, 74, 77, 114, and 110 respectively (as seen from Fig. 4(b)). Hence, Fig. 4 clearly shows the change of threat spectrum by time for Saudi Arabia upholding CR 2.

Requirement in CR 3 specifies that the proposed system should be able to predict cyber-attacks for any countries in the globe by analyzing past cyber-attack data. Fig. 5 shows cyber-attack is predicted for China on 9 November 2022 using the historical cyber-attack data from 11th October 2022 till 31 October 2022. The system presented in this paper allows a strategic decision maker to select any countries in the globe using the drop-down list and immediately the cyber threat is predicted by analyzing the past cyber-attack records upholding the requirement specified in CR 3. To validate the predicted cyber threat for China on 9 November 2023 (as forecasted to be 0.03 in Fig. 5), the actual threat statistics was obtained for that same day (i.e., 9 November 2023). Fig. 6 shows that the actual threat for China on 9 November 2023 was observed to be 0.03, which same as it was predicted to be in Fig. 5.

Requirement highlighted in CR 4 mandated detection of Anomalies in for any countries in the globe by analyzing past cyber-attack data. As



(a) Cyber threat spectrum for Saudi Arabia from 11 October 2022 and 31 October 2022



(b) Cyber threat spectrum for Saudi Arabia from 1 November 2022 and 8 March 2023

Fig. 4. Analyzing cyber threat spectrum against multiple dimensions of cyber-attacks between two different timeframes for Saudi Arabia satisfying CR 2.

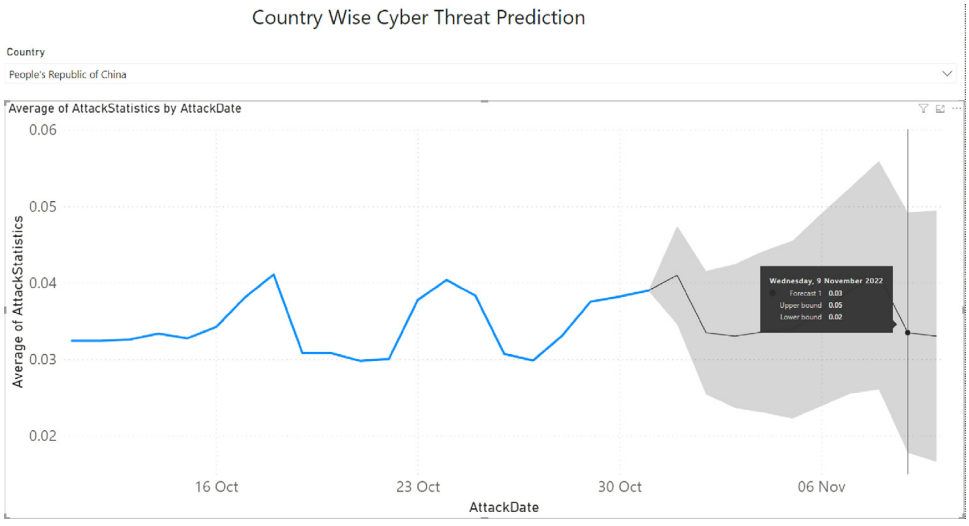


Fig. 5. Predict cyber-attacks for any countries in the globe by analyzing past cyber-attack data fulfilling CR 3.

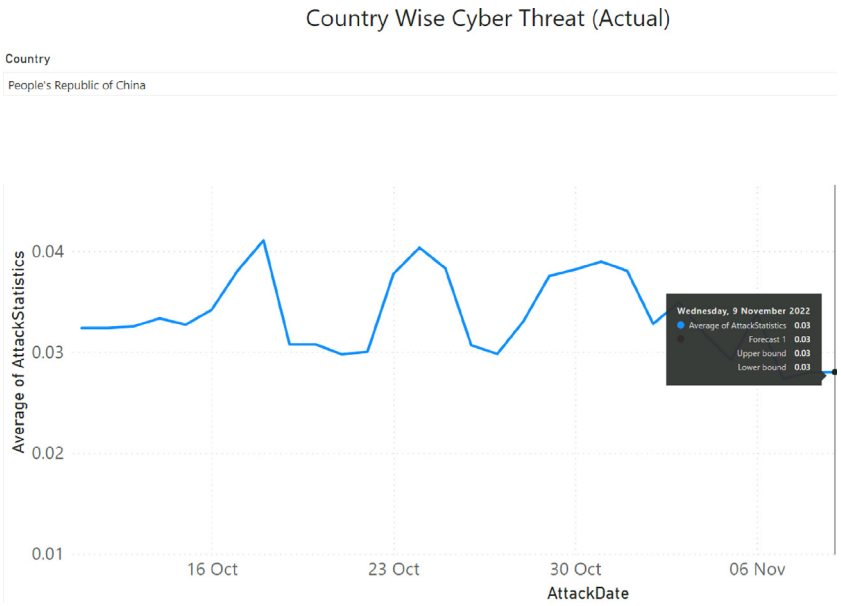


Fig. 6. Validation of predicted cyber-attack of Fig. 5 with actual cyber-attack data of 9 November 2022 for China (i.e., Validation of CR 3).

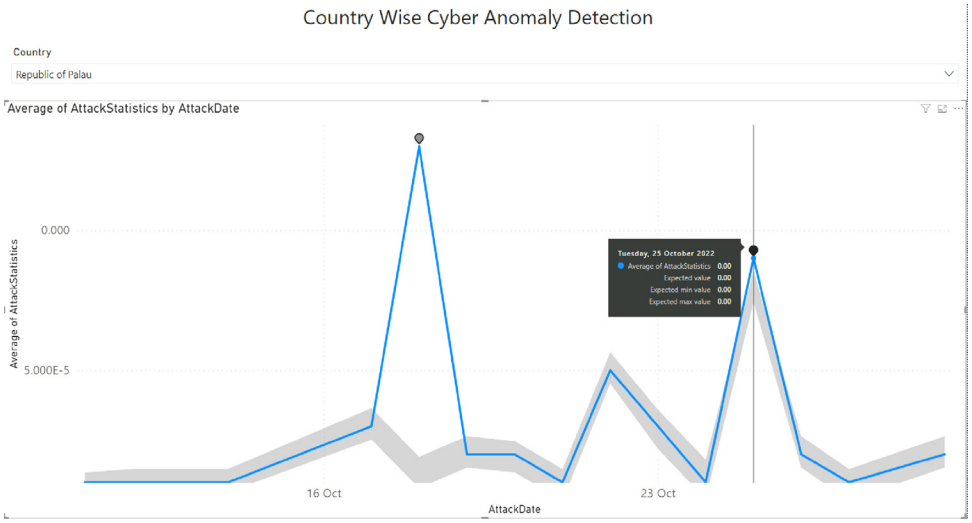


Fig. 7. Two Anomalies in cyber-threat spectrums for Republic of Palau was detected by analyzing past cyber-attack data satisfying CR 4.

seen from Fig. 7 Republic of Palau is selected from the top drop-down list and 2 anomalies were detected (one on the 18 October 2022 and the other one on the 25 October 2022) using CNN based deep learning. To validate the persistence of these 2 anomalies, anomaly detection was performed on the same country until end of 2022 (i.e., December 2022). As seen from Fig. 8, these 2 anomalies of Fig. 7, persisted until December 2022.

Fig. 9 demonstrates another feature of the proposed system, where a cyber-analyst is analyzing worldwide sentiments on cyber related topics with AI-based social media analysis. The user of the system can select any timeframe using the timeline, and immediately global sentiment during that specified time-duration is provided. The negative, positive, and neutral sentiment is represented with red, green, and gray colors. As seen from the pie chart in Fig. 9, the average sentiment of social media users between 11 October 2022 and 31 October 2022 were 36.62% negative, 21.05% positive, and 42.33%. The line chart just below the pie chart demonstrates the changing dynamics of user sentiments over time between 11 October 2022 and 30 October 2022. CR 5 mandates analyzing the view of social media users on various cyber issues over time. As seen from Fig. 7, Cyber related social media message in any language is obtained, followed by automated translation in English (for non-English posts) and Sentiment Analysis. This sentiment analysis process converts users' view into multiple sentiments (e.g., Positive, Negative, Mixed) as depicted in Fig. 9. It should be noted that within the context of CR 5, social media users' view is generic (i.e., anything related to "Cyber" or "Hack" keyword as shown previously in Fig. 1). Topic analysis had not been applied in this study. Thus, Fig. 9 demonstrates how the proposed system meets the mandates of CR 5.

Lastly, CR 6 requires the strategic users to perform analytical requirements from CR 1 to CR 5 on any devices (be it mobile, tablet, or desktops). While Figs. 3 to 9 demonstrated the proposed system running on Windows based desktop environment, Fig. 10 shows that worldwide cyber threat being analyzed within a tablet (i.e., Apple iPad) running iOS. Within the deployed iOS app, the decision-maker can clearly see that Bangladesh is the top-most threatened country by combining multiple dimensions of Cyber threats during the monitored period. After Bangladesh, the six most cyber threatened countries were

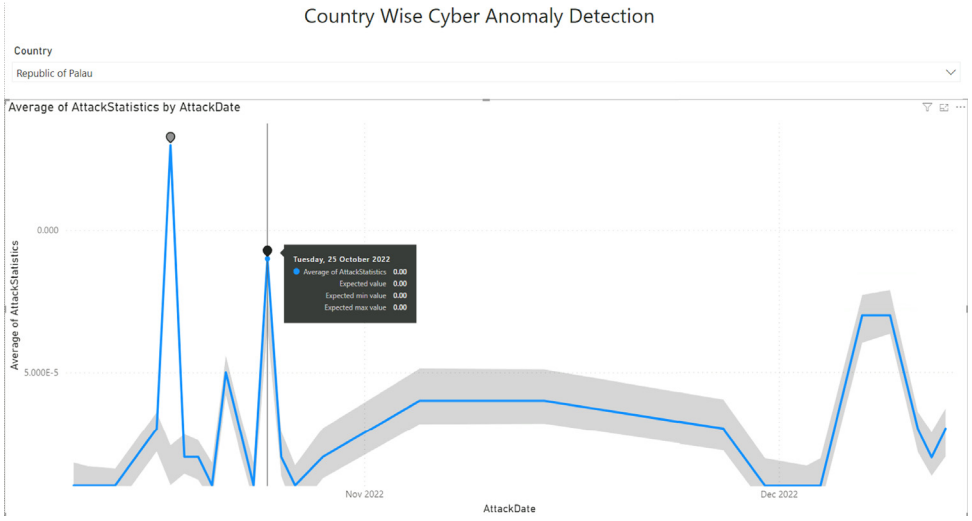


Fig. 8. Validation of identified anomalies across time (i.e., Validation of CR 4).

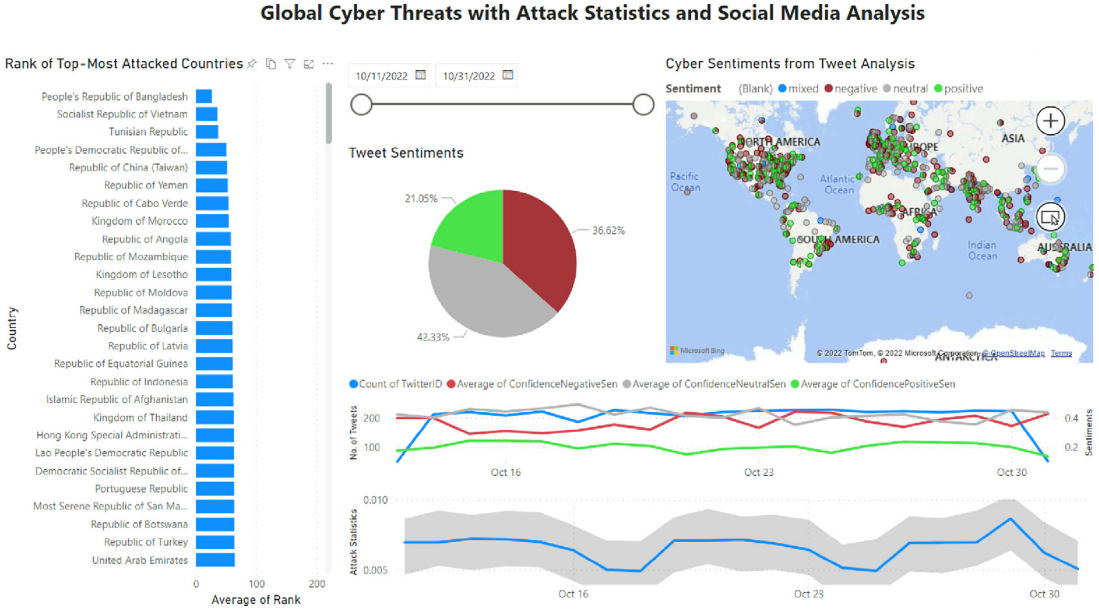


Fig. 9. Analyzing the view of social media users on various cyber issues between 11 October 2022 and 31 October 2022 upholding CR 5.

Vietnam, Tunisia, Algeria, China, and Yemen. This ranking changes with time. Using the information presented by our proposed solution, national cyber strategists can make informed decisions on the cyber posture of a country at any given time.

Finally, Fig. 11 demonstrates the deployed Android app running the presented solution within a mobile (i.e., Samsung Galaxy S23 Ultra). As it appears from Fig. 11, an analyst is conducting a detailed cyber threat analysis for India. During the monitored period India was mostly affected by Spam, Web threat, local infection, and other dimensions of cyber threats. Fig. 11 also demonstrated that during monitored period, India's spam rank was 9th in the world. The rank of web threats, exploits, malicious mails, ransomwares, spam, and network attack were 110, 81, 152, 108, 9, and 34 respectively.

Both Figs. 10 and 11 demonstrated how the requirement specified in CR 6 has been met by the proposed solution. At the beginning of this paper, 6 analytical requirements were identified. These critical requirements have not been addressed by existing studies in [11–19]. As seen from Table 5, these 6 critical requirements are completely fulfilled the proposed solution.

A demo version of the presented system is available at https://app.powerbi.com/view?r=eyJrIjoiYWJjOGY5YTUtZDBlNy00MTg1LWFkMTMtM2RmYzYzODQ1NzE1IiwidCI6IjBkMWI4YmRlLWZmYzEtNGY1Yy05NjAwLTJhNzUzZGFjYmEwNSJ9&pageName=ReportSection. As seen from Table 5, this scholarly paper presents a sophisticated cyber intelligence solution that comprises six critical decision criteria for evidence-based strategic decision-making in cybersecurity. Firstly, the solution entails an in-depth analysis of cyber threats, considering various dimensions of attacks for specific countries and comparing them globally (i.e., CR 1 of Table 5). This analysis provides valuable insights into the complexity of cyber threats and enables stakeholders to benchmark their cybersecurity measures against international counterparts, facilitating collaboration in the global fight against cybercrime.

Secondly, the solution incorporates real-time monitoring of threat fluctuations over time, enabling decision-makers to respond swiftly to emerging threats and allocate resources effectively (i.e., CR 2 of Table 5). By understanding temporal patterns in cyber-attacks, security teams can implement proactive defense measures and optimize cybersecurity budgets based on threat trends.

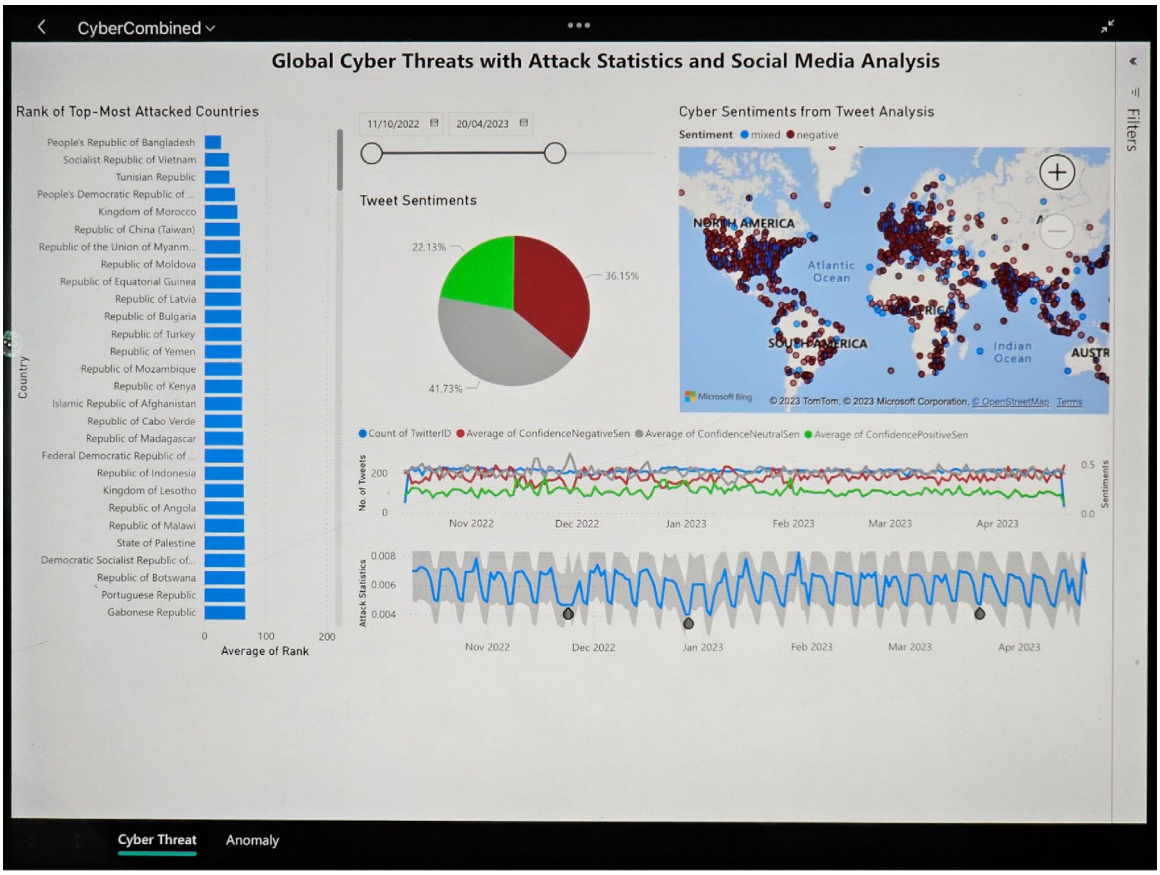


Fig. 10. Worldwide Cyber Threats being analyzed within deployed iOS App in Apple iPad 9th Generation (running within iOS Version 15) upholding CR 6.

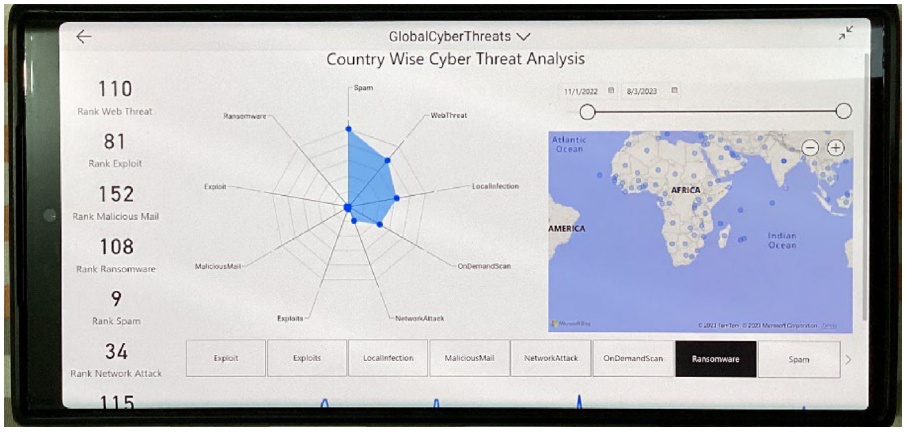


Fig. 11. Country Wise cyber threat analysis conducted for India with Samsung Galaxy S23 Ultra mobile phone running Android version upholding CR 6.

Table 5
Satisfying CR 1 to CR 6 by the proposed solution.

Critical requirements (CR)	Proposed solution meeting the CR
CR 1: Analyze threat spectrum against different dimensions of cyber-attacks for a particular country and compare with other countries in the globe	Fig. 3
CR 2: Analyze change of threat spectrum by time for any countries	Fig. 4
CR 3: Predict cyber-attacks for any countries in the globe by analyzing past cyber-attack data	Fig. 5, Fig. 6
CR 4: Detect Anomalies in cyber-threat spectrums for any countries in the globe by analyzing past cyber-attack data	Fig. 7, Fig. 8
CR 5: Analyze the view of social media users on various cyber issues over time	Fig. 8
CR 6: Obtain the above analytical requirements in any platforms on any devices	Fig. 3, Fig. 4, Fig. 5, Fig. 6, Fig. 7, Fig. 8, Fig. 9, Fig. 10, Fig. 11

Thirdly, the solution employs predictive analysis of past cyber-attack data to anticipate future threats (i.e., CR 3 of Table 5). This prognostic capability empowers organizations and governments to adopt preemptive measures, enhance resilience, and reduce the impact of potential cyber-attacks. The ability to predict and prepare for cyber threats is instrumental in safeguarding critical infrastructure and sensitive data.

Fourthly, the cyber intelligence solution is equipped with anomaly detection capabilities, which enable the identification of unusual patterns and targeted campaigns (i.e., CR 4 of Table 5). By detecting zero-day exploits and sophisticated attacks, security analysts can enhance incident response strategies, thereby minimizing potential damage.

Fifthly, the solution includes sentiment analysis of social media users' views on cyber issues, providing valuable insights into public awareness and concerns regarding cybersecurity (i.e., CR 5 of Table 5). Governments and cybersecurity organizations can leverage this analysis to tailor awareness programs and public engagement initiatives effectively.

Lastly, the solution's accessibility across multiple platforms and devices ensures widespread usability, enabling cybersecurity professionals and decision-makers to access critical threat intelligence anytime, anywhere, and make informed decisions irrespective of their location or preferred device (i.e., CR 6 of Table 5). This scalability contributes to a safer cyber landscape globally.

5. Conclusion

The prevailing cyber intelligence dashboards, as documented in reputable sources [11–19], are afflicted with numerous inadequacies. Such limitations encompass the dearth of country-level historical cyber-attack statistics, the incapacity to prognosticate country-specific cyber-threats, and inability of using CNN based anomaly detection methodologies on country-level cyber spectrums. In light of these deficiencies, the presented scholarly work introduces an avant-garde and ingenious solution that effectively rectifies all the aforementioned shortcomings inherent in existing cyber intelligence systems. The proposed solution furnishes an all-encompassing comprehension of cyber threats across the globe. Using a set of dashboards available in all platforms (iOS, Android, and Windows), a strategic decision maker can perform the following:

- Analyze and compare the cyber-threats among any number of countries in the world
- Analyze and compare the cyber-threat of a country by time
- Identify or detect anomalies within the cyber threat spectrum for any countries
- Predict cyber-threats for any countries
- Critically Analyze the views of Social Media users on cyber related issues

Using the presented system, a strategic cyber analyst would be able to analyze multidimensional cyber threats for any countries and accordingly recommend appropriate cyber posture. The overall prediction error using the actual data till 3 August 2023 was measured to be less than Root-mean-square deviation (RMSE) 0.19 using Eq. (9). This demonstrates relatively higher level of prediction accuracies compared to existing systems cyber threat prediction systems [54,56].

$$RMSE = \sqrt{\frac{\sum_{i=1}^{N} | \, |y(i) - \hat{y}(i)| \, |^2}{N}} \tag{9}$$

Recent research in AI-based Twitter discourse analytics has revealed that social media analysis encounters various challenges, including issues related to data quality, misinformation, fake user accounts, and ethical concerns [60,61]. In this study, it was assumed that all 30,203 tweets originated from genuine Twitter users' accounts. However, it is important to note that in real-time, tweets can be generated by fake users [62] and may contain misleading information [63].

Furthermore, when investigating the limitations of Social Media Analytics with a specific focus on Twitter, this study heavily relied on the use of real-time Tweet API, Microsoft Power Platform, and Microsoft Azure. Each of these platforms requires consistent financial investment, typically facilitated through credit card transactions. For instance, a basic subscription to the Twitter API, which grants access to a monthly quota of only 10,000 tweets, incurs a cost of $100 USD per month [64]. Expanding this allowance to cover 1 million tweets significantly increases the financial commitment to a substantial $5000 USD per month [64]. Due to these financial constraints, the research's scope was inevitably restricted, concentrating solely on a limited sample of tweets.

In future, we would like to develop innovative algorithms for generating country-wise cyber threat indexes purely from the social media data using more robust and modern toolset. For example, Microsoft recently launched Microsoft Fabric that is that encompasses robust and advanced data warehousing, data engineering, machine learning, data science, and visualization capability [65]. Using these innovative platforms, we envisage to use deep learning techniques like Recurrent Neural Network (RNN) [63], Auto-encoders and others. Since this study started to capture cyber related social media posts from 13 October 2022, creation of country-wise cyber threat index would be an extension to this research. Combining the social media driven cyber-threat indexes along with cyber-attack data from anti-virus vendors would provide much more comprehensive view of global cyber threats.

Declaration of competing interest

All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version. This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue. The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript.

Data availability

Data will be made available on request.

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